Jukes–Cantor Distance for the 6502life Noisy Copy Channel

6502life project

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Abstract

We derive closed-form evolutionary distance formulas for cells diverging through the 6502life noisy copy channel, where each bit is independently resampled with probability ε per copy event, and a resampled bit takes value 0 with probability q and value 1 with probability $1-q$. When $q = 1/2$ this is the standard binary Jukes–Cantor model; the general case covers erasure noise ($q = 1$) and arbitrary bias.

1 The Noisy Copy Channel

When a BRK noisy-copy fires (operands \$F5–\$F8), the origin cell's $M = 1024$ bytes are copied to a neighbor. Each bit is independently resampled with probability ε (`pBitNoise`), and faithfully copied with probability $1-\varepsilon$. When resampled, the bit becomes 0 with probability q (`pBitNoiseZero`) and 1 with probability $1-q$. The defaults are $\varepsilon = 1/2048$ and $q = 1/2$.

2 Biased Binary Jukes–Cantor Distance

Let $r = 1-\varepsilon$. Each bit evolves under a 2-state Markov chain with transition matrix

$$P = \begin{pmatrix} 1-\varepsilon(1-q) & \varepsilon(1-q) \\ \varepsilon q & 1-\varepsilon q \end{pmatrix}$$

with stationary distribution $\pi(0) = q$, $\pi(1) = 1-q$. The second eigenvalue is $r = 1-\varepsilon$.

Proposition 1. *After T copy events on the tree path between two cells whose ancestral bit distribution is at stationarity, the fraction of differing bits is*

$$p = 2q(1-q)\big(1-(1-\varepsilon)^T\big) \tag{1}$$

and the MLE of T given observed bit mismatch fraction $\hat{p}$ is

$$\boxed{\hat{T} = \frac{\log\big(1-\hat{p}/\big(2q(1-q)\big)\big)}{\log(1-\varepsilon)}} \tag{2}$$

Saturation: $p \to p_{\max} = 2q(1-q)$ as $T \to \infty$. When $q = 1/2$, $p_{\max} = 1/2$ and we recover the standard binary Jukes–Cantor formula $\hat{T} = \log(1-2\hat{p})/\log(1-\varepsilon)$.

Proof. Two bits, both starting in state s, evolve independently under P. After T steps the probability of being in state 0 given start state s is

$$P_T(0 \mid s=0) = q + (1-q)r^T, \qquad P_T(0 \mid s=1) = q(1-r^T).$$

Averaging over the stationary ancestor distribution $(\pi_0, \pi_1) = (q, 1-q)$, the probability that two independent descendants differ is

$$\begin{aligned} P(\text{diff}) &= \sum_{s \in \{0,1\}} \pi_s \cdot 2\, P_T(0 \mid s)\, P_T(1 \mid s) \\ &= 2q(1-q)(1-r^T). \end{aligned}$$

The MLE follows by solving for T. □

2.1 Byte-level distance

Since bits within a byte are independent, the probability that two corresponding bytes match is $(1-p)^8$, where p is the bit mismatch fraction:

$$P(\text{byte differs}) = 1 - (1-p)^8 \tag{3}$$

Inverting:

$$\hat{T} = \frac{\log\bigl(1 - \bigl(1 - (1-\hat{p}_{\text{byte}})^{1/8}\bigr) / \bigl(2q(1-q)\bigr)\bigr)}{\log(1-\varepsilon)} \tag{4}$$

2.2 Byte Hamming histogram

The per-byte Hamming distance $h \in \{0, \ldots, 8\}$ follows $\text{Bin}(8, p)$. The sufficient statistic is the sample mean $\bar{h}$, giving $\hat{p} = \bar{h}/8$. The histogram provides no additional information beyond the mean (it is a single-parameter exponential family), but can be used for goodness-of-fit testing against the copy-noise model.

2.3 Special cases

- $q = 1/2$: standard binary symmetric channel. Saturation at $p = 1/2$. This is the default.
- $q = 1$ (or $q = 0$): pure erasure noise. All resampled bits converge to the same value. Saturation mismatch $p_{\max} = 0$: two descendants become identical as $T \to \infty$. Distance estimation degenerates (the formula returns ∞ for any $\hat{p} > 0$).
- $q \neq 1/2$: biased noise. The equilibrium byte distribution is no longer uniform — bytes are biased toward $q > 1/2$ meaning more zero bits (sparser memory). This can model physical decay (UV erasure of EPROM, cosmic-ray bit clearing).

3 Applicability

The distance formula assumes all sequence divergence arises from the noisy copy channel. In practice, cross-contamination (random cells writing into each other's code regions) is often the dominant source of divergence and is not phylogenetically structured. The byte Hamming histogram can diagnose this: under the copy-noise model, $h \sim \text{Bin}(8, p)$; excess weight at $h \approx 4$ relative to $h = 0$ indicates non-copy contamination.